

The Position of System Ontology in the Theoretical Spectrum: A
Dialogue with Major Theoretical Frameworks

——A Position Paper

Author: Guorui He
Corresponding Author: Guorui He

Email: NooEcology@outlook.com

ORCID: 0009-0006-2947-0032

Abstract:

When readers first encounter *System Ontology*, *Noetic Ecology*, and *Civilization Dynamics*, they often react with a quick but misleading judgment: “Isn’t this just X?” This paper aims to dispel such misunderstandings by placing the core concepts of System Ontology side by side with nine representative theoretical frameworks, clearly showing similarities, differences, and original contributions. The frameworks engaged include: General System Theory, Dissipative Structure/Self-organization Theory, Autopoiesis Theory, Memetics, Kuhn’s Paradigm Theory, Dual-Process Theory, Integrated Information Theory, Predictive Processing/Free Energy Principle, as well as Embodied AI and Neuro-Symbolic AI (incorporating the Theorem of Consensual Manifestation of Intelligence). This paper does not attempt to prove the correctness of System Ontology; rather, it **positions** it within the academic landscape. A summary of eleven original contributions is provided at the end to help readers quickly grasp its uniqueness.

Keywords: System Ontology; Noetic Ecology; Position Paper; Theoretical Spectrum; Original Contributions

Statement of Relation to the Author’s Trilogy:

This paper is an independent positioning introduction to the three-volume work *System Ontology*, *Noetic Ecology*, and *Civilization Dynamics*. The trilogy maintains its internal self-consistency as an axiomatic-deductive system, while this paper serves as an external dialogue. It is not embedded in the trilogy but may be consulted before or after reading the trilogy.

Suggested Citation:

He, G. (2026). *The Position of System Ontology in the Theoretical Spectrum: A Dialogue with Major Theoretical Frameworks* [Position Paper]. GitHub.

License:

This work is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

1. Introduction

When readers first encounter *System Ontology*, *Noetic Ecology*, and *Civilization Dynamics*, they often react with a quick but understandable judgment: “Isn’t this just X?”

This response is entirely natural. After all, “self-organization tendency” sounds like Prigogine’s dissipative structures; “logical self-reference” evokes Hofstadter’s strange loops or Maturana’s autopoiesis; “noetic gene” is only one word away from Dawkins’s meme; “cognitive framework” closely resembles Kuhn’s paradigm. In academic reading, fitting new concepts into familiar categories is the most economical cognitive strategy.

However, this shortcut fails with System Ontology. Because it is not a collection of isolated concepts, but an **axiomatic-deductive system** — two meta-axioms at the foundation, seven system axioms unfolding from them, and dozens of theorems deduced, covering everything from elementary particles to civilizations. The similarity of terminology masks the distinctive **conceptual configuration, deductive structure, and dynamical mechanisms** of the system.

The task of this paper is not to prove the correctness of System Ontology — that is the work of the trilogy itself. The task is to **position** it: to tell readers where this theory stands in the academic landscape, what it shares with major frameworks, where it diverges, and what it has originally contributed.

To this end, this paper adopts the most straightforward method: **side-by-side comparison**. We bring System Ontology into dialogue with nine representative theoretical frameworks — General System Theory, Dissipative Structure/Self-organization Theory, Autopoiesis Theory, Memetics, Kuhn’s Paradigm Theory, Dual-Process Theory, Integrated Information Theory, Predictive Processing/Free Energy Principle, as well as Embodied AI and Neuro-Symbolic AI (incorporating the Theorem of Consensual Manifestation of Intelligence). For each dialogue, we provide a brief explanation and a comparative table, followed by a summary of eleven original contributions.

Reading advice: If time is limited, readers may directly consult the tables and the “Summary of Original Contributions” section. Those wishing to understand the differences in greater depth may read the paper sequentially.

This paper does not attempt to exhaust all relevant frameworks, nor does it seek to prove that System Ontology is superior to them. It merely provides a **map** — so that before entering the deductive world of the trilogy, readers know where they are.

2. Dialogue with General System Theory

General System Theory, founded by Ludwig von Bertalanffy in the mid-20th century, argues that there are universal laws of “systems” that cut across physics, biology, and society — such as wholeness, hierarchy, openness, and dynamic equilibrium. This idea provided a powerful philosophical weapon against reductionism and directly inspired subsequent complexity science.

System Ontology inherits the spirit of General System Theory’s “cross-disciplinary unification” ideal, but diverges on two fundamental issues.

First, the level of concern differs. Bertalanffy focused on “how systems operate” — how the whole becomes greater than the sum of its parts, how hierarchies are nested, how open systems maintain

homeostasis. This is **descriptive** systems science. System Ontology asks “why systems can exist at all” — why certain structures that emerge from potentiality can persist stably while others vanish instantly. This is **ontological** systems science.

Second, the degree of axiomatization differs. General System Theory provides a conceptual framework and heuristic principles, but does not formalize them into a system deducible from a few meta-axioms. System Ontology takes two meta-axioms (Self-organization Tendency and Systemic Constraint) as its foundation, deducing seven system axioms and dozens of theorems, forming a logically self-consistent deductive chain.

Third, the treatment of “constraint” differs. In General System Theory, boundaries and constraints are distinctions between system and environment, typically taken as externally given. System Ontology elevates “Systemic Constraint” to a meta-axiom dual to “Self-organization Tendency,” internalizing it as a mechanism of “legitimacy dissolution” for self-preservation — precisely the deep structure shared by cross-disciplinary phenomena such as Gödel’s incompleteness theorems, physical renormalization, and social institutional evolution.

Dimension	General System Theory (von Bertalanffy)	System Ontology
Core question	How do systems operate? (wholeness, hierarchy, openness)	Why can systems exist? (conditions of existence)
Self-organization	Mentioned as a feature of open systems	Meta-axiom 0 (the primary dynamic of being)
Constraint/boundary	System-environment distinction, externally given	Meta-axiom 1 (internalized self-constraint)
Methodology	Conceptual framework, heuristic principles	Axiomatic-deductive (meta-axioms → axioms → theorems)
Attitude toward “limitation”	Seen as boundary conditions of systems	Seen as necessary mechanism for “existential legitimacy”
Original difference	Descriptive systems science	Ontological dynamics + axiomatic deduction

Summary: System Ontology is not an “upgrade” of General System Theory, but an attempt to elevate systems science from a **descriptive** paradigm to an **ontological-deductive** paradigm. It answers not “what properties do systems have,” but “why systems can exist.”

3. Dialogue with Dissipative Structure/Self-organization Theory

Dissipative structure theory, proposed by Ilya Prigogine, argues that open systems far from equilibrium can spontaneously form spatiotemporal ordered structures (such as Bénard convection cells and

chemical oscillations) through continuous exchange of energy and matter with their environment. Stuart Kauffman and others further extended “self-organization” to biological systems, ecosystems, and even economic systems, emphasizing that order can emerge spontaneously without an external designer.

System Ontology shares with dissipative structure theory a fundamental concern with “self-organization,” but differs on three key points.

First, the level of explanation differs. Prigogine’s self-organization is a **thermodynamic mechanism**: it depends on physical conditions such as “far from equilibrium,” “energy dissipation,” and “nonlinear dynamics.” System Ontology’s self-organization tendency is an **ontological meta-axiom**: it is not any specific mechanism, but an inherent tendency of being itself, the ultimate driving cause for the local violation of the second law of thermodynamics (entropy increase). In other words, dissipative structure theory answers “how self-organization occurs,” while System Ontology answers “why self-organization is possible at all.”

Second, the role of constraint differs. Dissipative structure theory says almost nothing about “constraint” — it focuses on how order emerges from chaos, not on how emerged systems avoid self-destruction. System Ontology elevates “Systemic Constraint” to a meta-axiom dual to “Self-organization Tendency,” arguing that self-organization without constraint leads only to infinite hybridization or vicious self-reference, ultimately resulting in systemic collapse. The two together form the dynamical duality of being.

Third, the stance on teleology differs. Prigogine explicitly denies that self-organization has any purpose. System Ontology likewise denies any external, pre-established ultimate purpose, but holds that the self-organization tendency, in already-emerged systems, becomes **internalized as a “will to survive”** — the inherent tendency of a system to maintain its own existence. This “internalized teleology” is not metaphysical finalism, but a necessary consequence of system dynamics: a system without a self-preserving tendency simply could not exist stably.

Dimension	Dissipative	
	Structure/Self-organization Theory (Prigogine, Kauffman)	System Ontology
Status of self-organization	Thermodynamic mechanism (far-from-equilibrium + energy dissipation)	Ontological meta-axiom (a tendency of being itself)
Constraint	Not a core concept	Meta-axiom 1 (dual constraint principle)
Teleology	No purpose	No external purpose, but internalized as “will to survive”
Scope of explanation	Physical, chemical, some biological systems	Full scale: from elementary particles to civilizations
Explanation of collapse	Accidental or boundary conditions	Necessary phase transition when structural tension exceeds constraint

	Dissipative	
Dimension	Structure/Self-organization Theory (Prigogine, Kauffman)	System Ontology
		framework
Original difference	Mechanism level	Ontological level + dynamic duality + internalized will to survive

Summary: System Ontology elevates self-organization from a physical/chemical mechanism to an **ontological meta-axiom**, and supplements it with a dual “Systemic Constraint,” thereby explaining why self-organization does not expand infinitely into chaos but instead produces stable hierarchical structures. Dissipative structure theory describes how the “engine” runs; System Ontology asks why the engine exists and how it coordinates with its “brake.”

4. Dialogue with Autopoiesis Theory

Autopoiesis theory, developed by Humberto Maturana and Francisco Varela in the 1970s and later extended by Robert Rosen, centers on the concept of “autopoiesis” (self-production): a living system is a closed organization that continuously produces its own components through an internal network, thereby maintaining its boundary and identity. Autopoiesis theory emphasizes that the defining feature of living systems is **self-reference** and **operational closure** — the system takes itself as an object of operation and maintains its existence through this self-reference.

System Ontology shares with autopoiesis theory a fundamental concern with “self-reference,” but diverges on three fundamental issues.

First, the scope of self-reference differs. Autopoiesis theory strictly restricts self-reference to **living systems** — only systems capable of self-production possess genuine self-reference. System Ontology treats self-reference as a universal property of all complex systems (those reaching a specific threshold of complexity). From cellular gene regulatory networks, to brain neural networks, to social institutions, to artificial intelligence information processing systems — as long as the complexity condition is met, logical self-reference will emerge. This is a **bold generalization**, and precisely one of the most original claims of System Ontology.

Second, the functional role of self-reference differs. In autopoiesis theory, the core function of self-reference is **maintaining the system’s identity and boundary** — it is a “conservative” force. In System Ontology, logical self-reference serves as the **engine of innovation and driver of hierarchical transition**. Self-referential capacity enables a system to take itself as an object of cognition, constructing an internal “logical sub-universe” for simulation, iteration, and creative recombination. This is precisely the micro-dynamic source of paradigm revolutions, scientific discoveries, and technological breakthroughs.

Third, the relationship with constraint differs. Autopoiesis theory focuses on how systems maintain themselves through self-reference, but says little about “how systems avoid being destroyed by vicious self-reference.” System Ontology pairs self-reference with “Systemic Constraint” as a duality: logical

self-reference provides the driving force for innovation, while Systemic Constraint ensures that this innovation does not slide into infinite recursion or logical paradox. The “undecidable propositions” revealed by Gödel’s theorems are precisely the boundary drawn by Systemic Constraint in mathematical systems that possess strong self-referential capacity.

Dimension	Autopoiesis Theory (Maturana, Varela, Rosen)	System Ontology
Scope of self-reference	Restricted to living systems (self-production)	All complex systems reaching threshold complexity (from cells to civilizations)
Core function of self-reference	Maintaining system identity, operational closure (conservative)	Engine of innovation, driver of hierarchical transition (creative)
Handling of vicious self-reference	Not a central mechanism	Meta-axiom 1 (Systemic Constraint) dissolves its legitimacy
Relationship with constraint	Emphasizes closure, not duality	Self-reference and constraint form a duality, together constituting the dynamics of being
Interpretation of Gödel’s theorems	Not directly relevant	Seen as manifestation of Systemic Constraint in mathematics
Original difference	Biological paradigm	Generalization + innovation function + duality with constraint

Summary: Autopoiesis theory reveals how life maintains itself through self-reference; System Ontology reveals how self-reference drives innovation and evolution in all complex systems. The former answers “how to remain the same,” the latter answers “how to become different.” The two are not opposed but complementary — System Ontology extends the core insights of autopoiesis theory from biology to the entire spectrum of complex systems, while assigning self-reference a new creative role.

5. Dialogue with Memetics

Memetics, proposed by Richard Dawkins in *The Selfish Gene*, centers on the concept of the “meme”: the unit of cultural transmission, such as ideas, melodies, fashions, and technologies, which spread through imitation within a population. Like genes, memes undergo variation, selection, and heredity, driving cultural evolution.

System Ontology shares with memetics a concern for “cultural evolution,” but memetics remains at the level of **conceptual metaphor**, while System Ontology advances it to an **operationalizable scientific**

model.

First, the definition of the unit differs. The definition of the meme has long been vague: is a meme a single note, a song, an idea, or an ideology? Where are its boundaries? System Ontology’s “noetic gene” has a clear operational definition: symbols (concepts, narratives, cognitive frameworks) serve as the basic units of cultural heredity, variation, and selection. They replicate through education and transmission, vary through innovation and misreading, and are selected through social consensus and power structures. The noetic gene is anchored in symbolic systems, making it operationally tractable.

Second, the selection mechanism differs. Memetics lacks a clear explanation of “which memes are selected,” often appealing to “the meme’s own fitness” or “human preferences,” falling into circular reasoning. System Ontology specifies three selection mechanisms: (1) technical efficacy selection (in technological domains); (2) social resonance selection (in cultural domains); and (3) institutional filtering selection (in organizational domains). These selection mechanisms are independent of the memes themselves and are empirically testable.

Third, the dynamical model differs. Memetics provides no quantitative model of meme propagation. System Ontology provides the noetic gene propagation equation:

$$\frac{dP}{dt} = \beta \cdot I \cdot C \cdot P \cdot (1 - P) \cdot N - \delta P - \gamma PQ$$

where P is prevalence, β is infection intensity, C is framework compatibility, N is network effect coefficient, δ is forgetting rate, and γ is competition inhibition coefficient. This equation can simulate the rise, explosion, saturation, and decline of concepts, and has been semi-quantitatively validated with cases such as “involution.”

Fourth, the explanation of extreme altruism differs. Memetics explains extreme altruism as manipulation of the host by memes. System Ontology explains it uniformly as: an individual’s will to survive (Axiom 3) can extend its “self” boundary to internalized noetic entities (family, belief, civilization). Thus, sacrificing for a grand narrative is an expression of the will to survive operating at the scale of a larger noetic entity.

Dimension	Memetics (Dawkins)	System Ontology
Cultural unit	Meme (conceptual metaphor)	Noetic gene (operational definition: symbol)
Selection mechanism	Vague, circular	Clear: technical efficacy, social resonance, institutional filtering
Dynamics	No quantitative model	Propagation equation (simulable, testable)
Heredity mechanism	Imitation (vague)	Education, transmission, institutionalization
Explanation of extreme altruism	Meme manipulation of host	Expansion of “self” boundary of the will to survive
Original difference	Conceptual metaphor	Testable mathematical model + explicit mechanisms

Summary: Memetics opened the door to the study of cultural evolution, but the path beyond the door requires more precise tools. System Ontology, with its operational definition of the “noetic gene” and propagation equation, elevates memetics from a heuristic metaphor to a testable scientific hypothesis. This is not a denial of Dawkins, but an inheritance and deepening of his insights.

6. Dialogue with Kuhn’s Paradigm Theory

Thomas Kuhn, in *The Structure of Scientific Revolutions*, proposed paradigm theory, revolutionizing the cumulative picture of scientific progress. A paradigm is the set of theoretical frameworks, basic assumptions, and research methods shared by a scientific community. Normal science solves “puzzles” within the paradigm; when anomalies accumulate to a critical level, a paradigm revolution occurs, and a new paradigm replaces the old one. Kuhn’s work is descriptive and historically inductive — he tells us what happened in the history of science.

System Ontology shares with Kuhn a concern for “cognitive framework replacement,” but advances it from historical description to **dynamical mechanism**.

First, from description to mechanism. Kuhn described paradigm revolutions but did not explain why and when they occur. System Ontology’s “Cognitive Escape Theorem” provides a probabilistic formula for an individual’s cognitive framework transition:

$$P_{\text{escape}} = k \cdot \frac{M \cdot C}{A \cdot N}$$

where:

- P_{escape} : probability that an individual breaks through the current dominant cognitive framework per unit time
- M: individual’s metacognitive ability (proficiency in operating analytical lenses)
- C: intensity of cognitive dissonance (degree of conflict between perceived reality and internal framework)
- A: authority of the paradigm (institutionalization, logical self-consistency)
- N: number of individuals who have internalized the paradigm
- k: cognitive fluidity constant (degree to which civilizational infrastructure promotes cognitive mobility)

Note: In different documents, cognitive dissonance intensity may be denoted as D or C_d , with the same meaning. The system-level escape difficulty formula $D_c = \gamma \cdot \frac{A^2 \cdot N_i}{M_a}$ (where N_i is the proportion of individuals who have internalized the paradigm, and M_a is the average societal metacognitive ability) can be found in Theorem D23 of *Noetic Ecology* and is omitted here.

This formula transforms paradigm replacement from “historical contingency” into “computable probability,” revealing why pioneers are always few (when $A \cdot N$ is large, $P(\text{escape})$ is extremely small), and the levers for enhancing social innovation vitality (reduce $A \cdot N$, increase k and M).

Second, from qualitative to quantitative. Kuhn used “incommensurability” to describe the rupture between old and new paradigms, but provided no metric. System Ontology introduces the concept of “cognitive dimension” (DIM): dimensional transition ($D \rightarrow D+1$) is the geometric mechanism for breaking through cognitive involution. Within the same dimension, increasing knowledge (side length)

causes cognitive load to rise quadratically:

$$C_L = C_{\min} + \frac{A^2}{\text{DIM}^2}$$

where C_L is cognitive load, $C_{\min}$ is minimum cognitive cost, A is total knowledge, and DIM is the number of cognitive dimensions. Transitioning to a higher dimension causes cognitive load to plummet as the inverse square. This provides a mathematical definition of “involution”: cognitive heat death under dimensional lock-in.

Third, from history of science to all cognitive systems. Kuhn’s paradigm theory is confined to the history of science. System Ontology’s cognitive framework theory applies to all cognitive systems — from individual minds, organizational cultures, to civilizational paradigms. Religious reformations, artistic movements, corporate management changes, and personal worldview transformations all follow the same dynamical laws.

Dimension	Kuhn’s Paradigm Theory	System Ontology
Core question	How do paradigms change in the history of science? (descriptive)	How do cognitive frameworks transition? (mechanism)
Driving force	Accumulation of anomalies (qualitative)	Cognitive escape formula: $P = k \cdot MC / (AN)$ (individual level)
Nature of breakthrough	Paradigm shift (incommensurability)	Dimensional transition ($D \rightarrow D+1$)
Scope	History of science	All cognitive systems (individual, organization, civilization)
Explanation of “involution”	None	Cognitive heat death under dimensional lock-in
Original difference	Historical description	Computable dynamical formula + topological explanation

Summary: Kuhn provided a historical lens for understanding scientific progress. System Ontology provides a dynamical engine for understanding cognitive framework transitions in all cognitive systems. It answers the questions Kuhn left open: Why are paradigm revolutions so difficult? Because the probability of cognitive escape is inversely proportional to paradigm authority and the number of people who have internalized it. How to promote cognitive breakthroughs? Reduce authority monopolies, enhance metacognitive abilities, and increase cognitive fluidity.

7. Dialogue with Dual-Process Theory

Dual-process theory, systematically articulated by Daniel Kahneman in *Thinking, Fast and Slow*, has

become a cornerstone of behavioral economics and cognitive psychology. System 1 is fast, automatic, intuitive, emotional, and error-prone; System 2 is slow, controlled, analytical, rational, and reliable. Dual-process theory has successfully explained a wide range of cognitive biases (such as anchoring effects and availability heuristics), but its architecture has a fundamental omission.

System Ontology shares with dual-process theory the basic intuition of “cognitive layering,” but expands the two layers into a **three-tier architecture** and supplies the missing dimension.

First, the missing “biological directive layer.” In dual-process theory, System 1 is described as “intuition” and “emotion,” but the source of emotion and intuition — the organism’s survival monitoring system — is ignored. System Ontology clearly distinguishes: the **Biological Directive Layer** (based on evolutionarily shaped biological rules, outputting emotional signals), the **Subconscious Processing Layer** (distributed pattern recognition, outputting intuition), and the **Metacognitive Layer** (logical deduction and framework operation, outputting reason). Emotion and intuition are assigned to different layers, each with its own independent dynamical mechanisms.

Second, inter-layer relations as “synergistic resonance” rather than “competition.” In dual-process theory, System 1 and System 2 are often described as competing: reason must suppress intuition. System Ontology holds that healthy cognition is high-quality synergy among all three layers — the biological directive layer provides value anchoring and motivation, the subconscious processing layer provides efficient pattern recognition, and the metacognitive layer provides logical auditing and long-term planning. Excellent decision-making is not reason suppressing emotion, but high-alignment resonance between strong emotional motivation and strong rational auditing.

Third, provision of a quantitative tool: brain region alignment degree A. System Ontology directly links willpower intensity to “brain region alignment degree” A:

$$A = \frac{\sum_{i,j} w_{ij} \cdot \cos(v_i, v_j)}{\sum_{i,j} w_{ij}}$$

where v_i are the computational vectors of different brain functional regions. When A exceeds a critical threshold A_c , willed action is executed; when A falls below A_c , the system falls into cognitive dissonance. This provides a quantifiable dynamical explanation for subjective experiences such as willpower, decision confidence, and psychological conflict.

Dimension	Dual-Process Theory (Kahneman)	System Ontology
Number of layers	2 (System 1 / System 2)	3 (Biological Directive, Subconscious Processing, Metacognitive)
Location of emotion	Folded into System 1 (mixed with intuition)	Output of Biological Directive Layer (direct signal of will to survive)
Location of intuition	Folded into System 1 (mixed with emotion)	Output of Subconscious Processing Layer (anomaly report from pattern gestalt)
Location of	System 2	Function of Metacognitive Layer (logical

Dimension	Dual-Process Theory (Kahneman)	System Ontology
reason		deduction, framework operation)
Inter-layer relation	Competition / suppression	Synergistic resonance + brain region alignment degree A
Explanation of will	Not specified	Emergent global state when $A \geq A_c$
Original difference	Dual-process (qualitative)	Three-tier + alignment formula (quantifiable)

Summary: Dual-process theory revealed the duality of cognition, but conflated emotion with intuition and failed to explain the emergence of will. System Ontology expands the two layers into a three-tier architecture, supplies the missing “biological directive layer,” and uses “brain region alignment degree” to quantify the quality of emotion-reason synergy, thereby unifying the entire spectrum from impulsive decisions to high self-discipline.

8. Dialogue with Integrated Information Theory

Integrated Information Theory (IIT), proposed by Giulio Tononi, is one of the most influential formal theories in consciousness science. Its core claim is that consciousness equals the capacity of a system to integrate information, quantified by a mathematical value Φ (phi). The higher a system’s Φ , the higher its level of consciousness. Φ measures the degree to which a system as a whole is irreducible to the sum of its parts — that is, the system’s “integrated information.”

System Ontology shares with IIT the belief that consciousness is quantifiable, but the two diverge on fundamental orientations.

First, from “integration” to “synergy.” IIT focuses on the degree of “integration” of information within a system — whether the causal interactions among parts produce whole-system information that exceeds the sum of parts. System Ontology focuses on “synergistic resonance” across layers — the alignment among the Biological Directive Layer, the Subconscious Processing Layer, and the Metacognitive Layer. Integration is a static structural property; synergy is a dynamic process property. System Ontology holds that consciousness is not “how much integrated information a system possesses,” but the global state that emerges when the system operates with high alignment across its layers.

Second, the problem of content. One of the most significant criticisms of IIT is that it is “content-empty”: it can tell you whether a system is conscious and how conscious it is, but it cannot tell you what the content of consciousness is — why the experience is “the feeling of redness” rather than “the feeling of pain.” System Ontology’s “pattern gestalt” concept ($P = (S, R, C, W)$) provides a generative mechanism for conscious content: conscious content arises from the resonance of the three layers around a specific pattern gestalt, with its rich qualia jointly constituted by the sensory emotions

(S) of the Biological Directive Layer, the pattern details (R) of the Subconscious Processing Layer, and the symbolic narratives (C) and value weights (W) of the Metacognitive Layer.

Third, handling of counterintuitive predictions. IIT leads to counterintuitive conclusions, such as that a highly connected grid or photodiode array might have a high Φ and thus be considered conscious. System Ontology rules this out through the “will to survive” (Axiom 3) and “value criterion” (Axiom 5): consciousness is not pure information integration, but a functional architecture serving the survival and development of the system. A system without a will to survive has no evolutionary reason or functional necessity to be conscious, even if it has high integrated information.

Fourth, testability. Computing Φ for complex systems is extremely difficult (combinatorial explosion); in practice, it can only be calculated for very small systems. System Ontology’s “alignment degree A” is based on the similarity of computational vectors across brain functional regions, and can in principle be measured using existing technologies such as fMRI and EEG to assess cross-layer neural synchrony, making it more directly empirically testable.

Dimension	Integrated Information Theory (Tononi)	System Ontology
Measure of consciousness	Φ (degree of information integration)	Alignment degree A (cross-layer synergistic resonance)
Nature of measure	Static structural property	Dynamic process property
Explanation of content	None (content-empty)	Pattern gestalt (S,R,C,W) generates content
Counterintuitive predictions	High- Φ systems may be conscious (e.g., grids)	Ruled out: requires will to survive and value criterion
Testability	Computationally difficult, limited to small systems	In principle measurable with fMRI/EEG
Theoretical orientation	Information-centered	Function-centered (serving survival and development)
Original difference	Mathematical quantity Φ	Functional architecture + content generation mechanism + testability

Summary: IIT captures a key dimension of consciousness — information integration — with mathematical elegance. System Ontology attempts to answer the questions IIT cannot: Why does consciousness have content? Why does consciousness serve survival? How does consciousness emerge from a layered architecture? It is not a denial of IIT, but a complementary approach that shifts from an “information-centered” to a “functional-synergistic” perspective, offering a different path toward a quantifiable science of consciousness.

9. Dialogue with Predictive Processing/Free Energy Principle

Predictive processing, particularly as formalized by Karl Friston's Free Energy Principle, is one of the most influential unifying theories in contemporary cognitive science and neuroscience. Its core claim is that the brain is a hierarchical prediction machine, continuously generating predictions about the causes of the world and minimizing prediction error (free energy) by updating internal models or acting on the environment. Consciousness, within this framework, is typically interpreted as high-level hypotheses under active optimization with high uncertainty.

System Ontology is deeply compatible with predictive processing, but situates it as **one layer within a larger architecture** and supplies its missing value anchor.

First, locating predictive processing as the core algorithm of the “Subconscious Processing Layer.” Predictive processing elegantly describes how the brain efficiently perceives, learns, and acts — but this is precisely the operating mechanism of the “Subconscious Processing Layer” in System Ontology. Predictive processing answers “how the system processes information efficiently,” but does not answer “why the system processes this information rather than that” or “how much processing is enough.” System Ontology places predictive processing in the middle of a three-tier architecture, with the Biological Directive Layer providing the value anchor above it and the Metacognitive Layer providing logical auditing alongside or above it.

Second, supplying the missing value anchor. The core of the Free Energy Principle is “minimizing prediction error,” but this principle has no intrinsic value content: a system could minimize error by avoiding uncertainty (leading to rigidity) or by distorting perception to match expectations (leading to hallucination). System Ontology's Biological Directive Layer provides the ultimate criterion — the will to survive. What the system seeks to minimize is not abstract prediction error, but “survival disparity” — threats to its continued existence. Prediction errors are prioritized only when they relate to survival. This answers the fundamental question: “Why is this error more important than that error?”

Third, reinterpreting “active inference.” The Free Energy Principle's “active inference” states that a system can minimize error not only by updating its internal model but also by changing the environment to make sensory inputs conform to expectations. System Ontology interprets this as the embodied expression of the will to survive: the system acts to reshape the environment to be more favorable to its own persistence. This is not a denial of the Free Energy Principle, but a provision of the ultimate “why” for action.

Fourth, a unified explanation of psychopathology. Predictive processing explains depression, anxiety, and other mental disorders as dysfunctions of prediction coding (e.g., overly precise or overly imprecise priors). System Ontology adds another dimension: mental disorders are fundamentally breakdowns of synergy among the three layers of consciousness. For example, depression is not merely a disorder of prediction error processing, but a systemic alignment failure among the Biological Directive Layer (low-valence emotional signals), the Subconscious Processing Layer (entrenched negative patterns), and the Metacognitive Layer (inability to reconstruct meaningful narratives). This provides richer targets for psychotherapeutic intervention.

Dimension	Predictive Processing/Free Energy Principle (Friston)	System Ontology
-----------	--	-----------------

Dimension	Predictive Processing/Free Energy Principle (Friston)	System Ontology
Core principle	Minimizing prediction error (free energy)	Layer synergy driven by will to survive
Value anchor	None (error minimization has no intrinsic direction)	Biological Directive Layer provides will to survive as ultimate criterion
Position in architecture	Attempts to explain everything (global theory)	Core algorithm of the Subconscious Processing Layer (mid-level mechanism)
Explanation of action	Active inference: change environment to match predictions	Embodiment of will to survive: reshape environment for persistence
Psychopathology	Predictive coding dysfunction	Three-layer synergy breakdown + predictive coding dysfunction
Relation to metacognition	Not specified	Metacognitive Layer can monitor and correct predictive coding
Original difference	Algorithmic level (elegant, but lacking teleology)	Situates FEP as mid-level mechanism, supplies value anchor and layer architecture

Summary: Predictive processing is an excellent theory for understanding the brain's information-processing mechanisms. System Ontology places it within a larger three-tier architecture, uses the “will to survive” to supply its missing value anchor, and extends its dynamics with the Biological Directive Layer above and the Metacognitive Layer alongside. The two are not opposed — predictive processing describes how the “engine” runs efficiently; System Ontology describes why the engine exists, whom it serves, and how it coordinates with higher-level control systems.

10. Dialogue with Embodied AI and Neuro-Symbolic AI (Including the Theorem of Consensual Manifestation of Intelligence)

Embodied AI argues that intelligence must emerge through real-time interaction between a physical body and the environment; meaning must be “grounded” in action. Neuro-Symbolic AI attempts to combine the perceptual learning capabilities of neural networks with the logical reasoning capabilities of symbolic systems, in order to overcome the black-box nature of pure neural networks and the fragility of pure symbolic systems. These two paradigms represent major contemporary approaches to artificial intelligence research.

System Ontology neither denies the importance of physical interaction nor rejects the necessity of

symbolic reasoning, but provides a more fundamental perspective: the **Theorem of Consensual Manifestation of Intelligence** (Theorem F31 of *Noetic Ecology*).

The Theorem of Consensual Manifestation of Intelligence states: Intelligence is not an inherent property of any entity, but rather a goal-directed rational landscape realized when a logically self-consistent axiomatic system (cognitive framework) is loaded into a medium with information-processing potential and activated through a closed loop of perception, action, and feedback with the external world. Human intelligence and artificial intelligence are isomorphic implementations of the same rational structure in different media (carbon-based / silicon-based); their similarity arises from the cross-media reproducibility of the logical structure of the axiomatic system, not from any intrinsic property of the medium.

Formal expression:

$$I = C \cdot \bigcap_P R(A, P)$$

where:

- I : Intelligence
- C : Consensual degree (intersection of consensus across processors)
- A : Axiomatic system (cognitive framework)
- P : Processor (human, AI, group, agent)
- R(A, P) : Rational landscape output by the axiomatic system when running on the processor

Based on this theorem, System Ontology responds to Embodied AI and Neuro-Symbolic AI as follows:

First, embodiment is a sufficient condition, not a necessary one. Physical interaction is one path to intelligence (the carbon-based path), but not the only path. As long as a system can load the same axiomatic system and possesses an effective closed loop for interacting with the external world (not necessarily a physical body — sensors, data interfaces, or even simulated environments suffice), it can exhibit rationally intelligent behavior isomorphic in structure. Insisting on replicating the human body is like trying to perfect the feathers of a bird to achieve flight while ignoring the underlying principles of aerodynamics.

Second, Neuro-Symbolic AI is on the right track, but requires a unifying axiomatic system. The core problem facing Neuro-Symbolic AI is the “binding problem” — how to seamlessly connect the black-box outputs of neural networks with the precise reasoning of symbolic systems. System Ontology’s answer is a **shared axiomatic system**. The neural network extracts patterns from experience (function of the Subconscious Processing Layer), while the symbolic system performs logical verification and deduction based on the axioms (function of the Metacognitive Layer). Their “binding” is not a patchwork engineering solution but a cognitive isomorphism achieved by loading the same axiomatic system.

Third, the nature of intelligence is consensual manifestation, not an inherent property of any medium. Both Embodied AI and Neuro-Symbolic AI implicitly assume that intelligence is tied to a specific medium. System Ontology denies this: the same axiomatic system can be activated in different media (human brain, silicon chips, quantum computers, or even social networks), producing functionally equivalent rational intelligence. This means that AI need not “be like a human”; it only needs to share the same fundamental rational axioms with humans to achieve deep cognitive alignment.

Dimension	Embodied AI	Neuro-Symbolic AI	System Ontology (Consensual Manifestation of Intelligence)
Source of intelligence	Real-time interaction between body and environment	Neural network perception + symbolic logical reasoning	Activation of axiomatic system in a medium + consensus
Core problem	How is meaning grounded?	Binding problem between symbols and perception	Cross-medium isomorphism of rational structure
Necessity of body	Necessary (sufficient condition)	Not necessary	Sufficient but not necessary
Necessity of symbols	Not emphasized	Necessary	Necessary, but symbols must be anchored to axioms
Medium specificity	Yes (biological body)	No (but not explicitly thematized)	No (axiomatic systems are cross-media reproducible)
Nature of intelligence	Behavioral patterns	Combination of perception + reasoning	$I = C \cap R(A, P)$ (consensual manifestation)
Implication for AI alignment	Needs a body	Needs symbols	Needs axiomatic system + consensus protocols
Original difference	Intelligence = bodily experience	Intelligence = neural + symbolic	Intelligence = consensual activation of axiomatic system in a medium

Summary: Embodied AI and Neuro-Symbolic AI each capture one aspect of intelligence, but neither touches its essence — the consensual manifestation of an axiomatic system in a medium. System Ontology’s theory of intelligence transcends the “body-necessary” vs. “symbol-necessary” debate, showing that any medium capable of loading the same axiomatic system and interacting effectively with the world through a closed loop can instantiate isomorphic rational intelligence. This perspective provides a new theoretical foundation for general artificial intelligence: not imitating the human brain or body, but discovering and instantiating more powerful logical frameworks.

11. Summary of Original Contributions

Through the dialogues with nine theoretical frameworks above, the distinctive contours of System Ontology have emerged. It is neither a rejection of any school nor a simple patchwork of concepts from multiple schools. Rather, through **axiomatic-deductive** method, it recombines existing concepts and endows them with new dynamical mechanisms, forming a novel system with internal logical consistency.

This section lists the original contributions concisely, helping readers quickly grasp what makes System Ontology unique. Each contribution is summarized in one sentence, with its primary source in the trilogy indicated.

1. Meta-axiom Duality: The Co-founding of Self-organization Tendency and Systemic Constraint

The “Self-organization Tendency” (dynamic) and “Systemic Constraint” (constraint) are posited as dual meta-axioms, together constituting the dynamical foundation of being. Existing theories emphasize either self-organization (Prigogine, Kauffman) or constraint (cybernetics, institutional theory); none have taken the two as dual meta-axioms simultaneously grounding ontology.

Source: System Ontology, Meta-axiom 0 and Meta-axiom 1

2. Generalization of Logical Self-reference: From Life to All Complex Systems

Self-reference is extended from a property of living systems (autopoiesis theory) to a universal property of all complex systems that reach a specific threshold of complexity, and is assigned a new role as the “engine of innovation” and “driver of hierarchical transition.” This generalization, though controversial, is one of System Ontology’s most original claims.

Source: System Ontology, Axiom 1

3. Cross-layer Unification of Duality: Matter/Noetic, Hierarchy/Substrate, System/Rules

The concept of “duality” is used at three different levels (matter/noetic, hierarchy/substrate, system/rules) and unified within a single theoretical framework to explain emergent relations from physics to society. Existing theories discuss “dualism” or “duality,” but none have employed it as a cross-layer unifying principle.

Source: System Ontology, Axiom 2 and Axiom 6

4. Operationalization of the Noetic Gene: Symbols as Units of Cultural Evolution + Propagation Equation

“Symbols” are explicitly defined as the basic units of cultural heredity, variation, and selection, and a quantifiable propagation equation is provided: $\frac{dP}{dt} = \beta \cdot I \cdot C \cdot P \cdot (1 - P) \cdot N - \delta P - \gamma PQ$. Memetics

(Dawkins) provided a conceptual metaphor but lacked operational definitions and dynamical models.

Source: Noetic Ecology, Axiom 5; Civilization Dynamics, Chapter 2

5. Three-Tier Model of Consciousness + Brain Region Alignment Degree Formula

Consciousness is deconstructed into the Biological Directive Layer, Subconscious Processing Layer, and Metacognitive Layer, with a weighted cosine similarity formula quantifying the “brain region alignment degree” A , establishing a positive correlation between willpower intensity and A . Dual-process theory (Kahneman) has only two layers and conflates emotion with intuition; Integrated Information Theory (Tononi)’s Φ cannot explain the content of consciousness.

Source: Noetic Ecology, Theorem E25; The Synergistic Resonance Model of Consciousness

6. Axiom-Trimming Hypothesis: The “Alignment Problem” of the Human Mind

The framework of the AI alignment problem is applied inversely to the human mind, proposing that cognitive biases arise from subconscious pre-training weights (experience) not being trimmed by the axiomatic system (logical rules). Existing cognitive psychology (dual-process theory) has no concept of “trimming.”

Source: When a Stone Is Not a Tool

7. Cognitive Escape Theorem + Dimensional Criterion

A probabilistic formula for an individual’s cognitive framework transition is given:

$$P_{\text{escape}} = k \cdot \frac{M \cdot C}{A \cdot N}$$

where:

- P(escape): probability of breaking through the current dominant cognitive framework per unit time
- M: individual’s metacognitive ability
- C: intensity of cognitive dissonance
- A: authority of the paradigm
- N: number of individuals who have internalized the paradigm
- k: cognitive fluidity constant

Note: In different documents, cognitive dissonance intensity may be denoted as D or C_d, with the same meaning. The system-level escape difficulty formula $D_c = \gamma \cdot \frac{A^2 \cdot N_i}{M_a}$ (where N_i is the proportion of individuals who have internalized the paradigm, and M_a is the average societal metacognitive ability) can be found in Theorem D23 of Noetic Ecology.

Simultaneously, dimensional transition ($D \rightarrow D+1$) is proposed as the geometric mechanism for breaking through cognitive involution, providing a mathematical definition of “involution”: within the same dimension, increasing knowledge causes cognitive load to rise quadratically, $C_L = C_{\min} + \frac{A^2}{\text{DIM}^2}$; transitioning to a higher dimension causes cognitive load to plummet as the inverse square. Kuhn’s paradigm theory is descriptive, lacking dynamical formulas and topological explanations.

Source: Noetic Ecology, Theorem D23 and D24

8. Three Criteria for Identifying Systemic Parasites

Three criteria — responsibility evasion, rule distortion, and value destruction — are proposed for diagnosing subsystem alienation. Existing sociological theories (e.g., Merton’s “dysfunction”) lack such operationalized diagnostic tools.

Source: Noetic Ecology, Theorem D21

9. Civilization State Function

The formula is proposed:

$$\text{文明潜能} = \Sigma(\text{个体生态位实现度}) \times \text{系统资源流动性} \times (1 - \text{系统寄生体负担})$$

This quantifies civilizational health into three intervenable factors. Existing civilizational studies (Toynbee, Huntington) have no such quantitative criterion.

Source: Civilization Dynamics, Chapter 3

10. Conflict-Driven Parliamentary AGI Architecture

It is proposed that cognitive conflict be internalized as a system driving force, achieving endogenous alignment through multi-agent debate and dynamic concept library evolution. Existing AI alignment methods (RLHF, Constitutional AI) are all “external correction.”

Source: The Logical Sub-universe Parliament

11. Consensual Manifestation of Intelligence

Intelligence is proposed not as an inherent property of any medium, but as the consensual activation of an axiomatic system in a medium: $I = C \cdot \bigcap_p R(A, P)$. Human intelligence and artificial intelligence are isomorphic implementations of the same rational structure in different carriers. Embodied AI emphasizes the necessity of a body; Neuro-Symbolic AI emphasizes perception + reasoning combination; this theory unifies them under the cross-media reproducibility of axiomatic systems.

Source: Noetic Ecology, Theorem F31

Summary

Among the eleven original contributions above, some are at the ontological level (meta-axiom duality, generalization of self-reference, cross-layer unification of duality); some are quantifiable mathematical models (propagation equation, alignment degree formula, cognitive escape formula, civilization state function); some are operationalized diagnostic tools (axiom-trimming hypothesis, three criteria for systemic parasites); some are architectural designs for AI (conflict-driven parliamentary AGI); and one redefines the nature of intelligence itself (consensual manifestation). Together, they constitute the distinctive value of System Ontology — not the provision of a few new concepts, but a **new axiomatic-deductive framework** and a **series of testable, operational theoretical tools** derived from it.

12. Conclusion

This paper began with a simple observation: readers, encountering similar terminology, often react with the misunderstanding, “Isn’t this just X?” This misunderstanding is not the reader’s fault, but a natural cognitive inertia in academic reading. The mission of this paper has been to dispel this misunderstanding — not by proclaiming “my theory is better,” but through **side-by-side comparison**, clearly showing the similarities, differences, and original contributions between System Ontology and nine representative theoretical frameworks.

Looking back over the whole paper, a clear picture emerges:

First, System Ontology is not a rejection of existing schools. It acknowledges the holistic insights of General System Theory, the revelation of self-organization by Dissipative Structure Theory, the emphasis on self-reference by Autopoiesis Theory, the heuristic of cultural evolution by Memetics, the description of paradigm change by Kuhn, the intuition of cognitive layering by Dual-Process Theory, the attempt at quantification by Integrated Information Theory, the elucidation of

information-processing mechanisms by Predictive Processing, and the explorations of intelligence by Embodied AI and Neuro-Symbolic AI. System Ontology stands on the shoulders of these giants.

Second, System Ontology is not a simple patchwork of existing concepts. Its originality lies not in inventing entirely new “atomic concepts,” but in recombining these concepts into a new axiomatic-deductive framework and endowing them with new dynamical mechanisms. Self-organization is elevated from a thermodynamic mechanism to an ontological meta-axiom; self-reference is extended from a property of life to all complex systems; duality is expanded from a single-level phenomenon to a cross-layer unifying principle; memes are advanced from metaphor to operational noetic genes; paradigm change is transformed from historical description to a computable cognitive escape formula. These are not “renaming,” but **new combinations, new positions, and new deductions of concepts.**

Third, the ultimate test of System Ontology lies not in this paper but in the trilogy and its applications. This paper is a positioning, not a defense. The real defense lies in: Is the axiomatic-deductive system of *System Ontology* self-consistent? Can the theorems of *Noetic Ecology* explain cognitive, cultural, and social phenomena? Can the equations of *Civilization Dynamics* diagnose civilizational states and predict evolutionary trajectories? And are the applications in artificial intelligence, psychology, and education effective? The answers to these questions require reading and practicing the trilogy.

Fourth, an invitation. If you are a reader encountering System Ontology for the first time, I hope the comparative tables and summary of original contributions in this paper will help you quickly orient yourself, avoiding time wasted in the “isn’t this just X” misunderstanding. If you are already familiar with the trilogy, I hope this paper provides an “external map” for you — so that before entering the deductive castle, you can see its relation to the surrounding landscape.

System Ontology does not claim to be absolute truth. It is merely a new cognitive tool — a scalpel forged from first principles. It may prove sharp, or it may prove flawed. But in either case, its value lies in the **dialogue it opens**, not in the **debates it ends**. This paper is the first “positioning guide” for that dialogue.

13. References

Part I: Core Literature of the Dialogued Schools

General System Theory

1. von Bertalanffy, L. (1968). *General System Theory: Foundations, Development, Applications*. New York: George Braziller.

Dissipative Structure / Self-organization Theory

2. Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos: Man's New Dialogue with Nature*. New York: Bantam Books.

3. Kauffman, S. A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford: Oxford University Press.

Autopoiesis Theory

4. Maturana, H. R., & Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. Boston: D. Reidel.

5. Rosen, R. (1991). *Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life*. New York: Columbia University Press.

Memetics

6. Dawkins, R. (1976). *The Selfish Gene*. Oxford: Oxford University Press.

7. Blackmore, S. (1999). *The Meme Machine*. Oxford: Oxford University Press.

Paradigm Theory

8. Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.

Dual-Process Theory

9. Kahneman, D. (2011). *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.

Integrated Information Theory

10. Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1), 42.

11. Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450–461.

Predictive Processing / Free Energy Principle

12. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

13. Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.

Embodied AI and Neuro-Symbolic AI

14. Anderson, M. L. (2003). Embodied cognition: A field guide. *Artificial Intelligence*, 149(1), 91–130.

15. Garcez, A. d., & Lamb, L. C. (2020). Neurosymbolic AI: The 3rd wave. *arXiv preprint arXiv:2012.05876*.

16. Sarker, M. K., Zhou, L., Eberhart, A., & Hitzler, P. (2021). Neuro-symbolic AI: an emerging field with promising prospects. *AI Magazine*, 42(4), 14–28.

Part II: Relevant Works by the Author

17. He, G. (2026). *System Ontology: The Ultimate Jurisprudence of Being*. GitHub. <https://github.com/wenhe1994/wenhe001>

18. He, G. (2026). *Noetic Ecology: A Systems Theory of Cognition, Culture, and Society*. GitHub. <https://github.com/wenhe1994/wenhe001>
19. He, G. (2026). *Civilization Dynamics: Planetary Civilization Studies*. GitHub. <https://github.com/wenhe1994/wenhe001>
20. He, G. (2026). *Introduction to Cognitive Protocols: From Disciplinary Consensus to Systems-Theoretic Foundations*. GitHub. <https://github.com/wenhe1994/wenhe001>
21. He, G. (2026). *The Two Laws of Being: An Ontological Framework of Self-organization Tendency and Systemic Constraint*. Preprint. GitHub.
22. He, G. (2026). *A New Interpretation of Gödel's Incompleteness Theorems: As a Proof of Existential Legitimacy through Systemic Constraint*. Preprint. GitHub.
23. He, G. (2026). *The Synergistic Resonance Model of Consciousness: Hierarchical Integration of Biological Drive, Subconscious Processing, and Metacognitive Control*. Preprint. GitHub.
24. He, G. (2026). *When a Stone Is Not a Tool: A Case Study from a Comment Section Revealing the Human Mind's Alignment Problem*. Preprint. GitHub.
25. He, G. (2026). *From Probability Machine to Axiomatic Machine: Why Large Language Models Cannot Understand World Rules*. Preprint. GitHub.
26. He, G. (2026). *A Systems-Dynamics Architecture for AGI Alignment: The Logical Sub-universe Parliament and Dynamic Concept Evolution*. Preprint. GitHub.